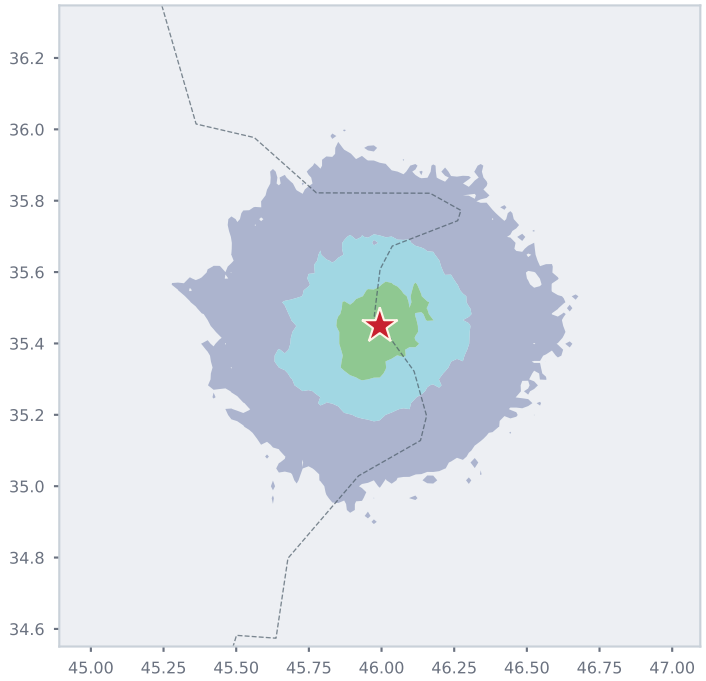
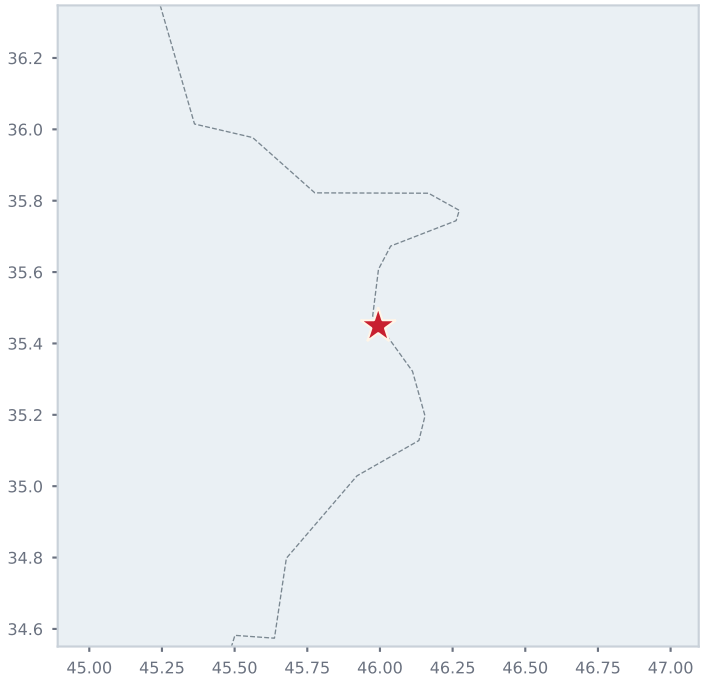
Shaking intensity (IMS-25)



Expected building damage grade



IMS-25 intensity



Expected damage grade (IMS-25)



Damage alert

GREEN Little or no building damage expected

Provisional **Day occupancy**

Buildings heavily damaged (DG3 or worse)

18 to 300

Central estimate 70. The range is the 5th to 95th percentile.

Buildings in the shaken area

618 thousand

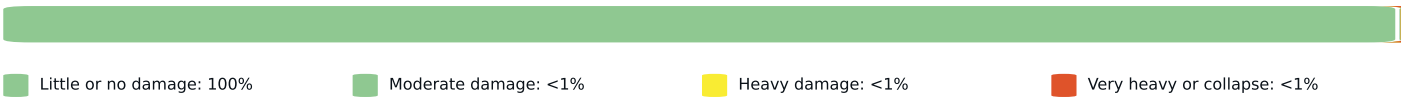
21% of them inferred from built-up area, not observed footprints.

People by shaking level

V	53 thousand
IV	204 thousand

Building exposure modelled for: Iraq. Shaded areas without exposure data are shown for shaking only.

Share of buildings in the mapped area by expected damage

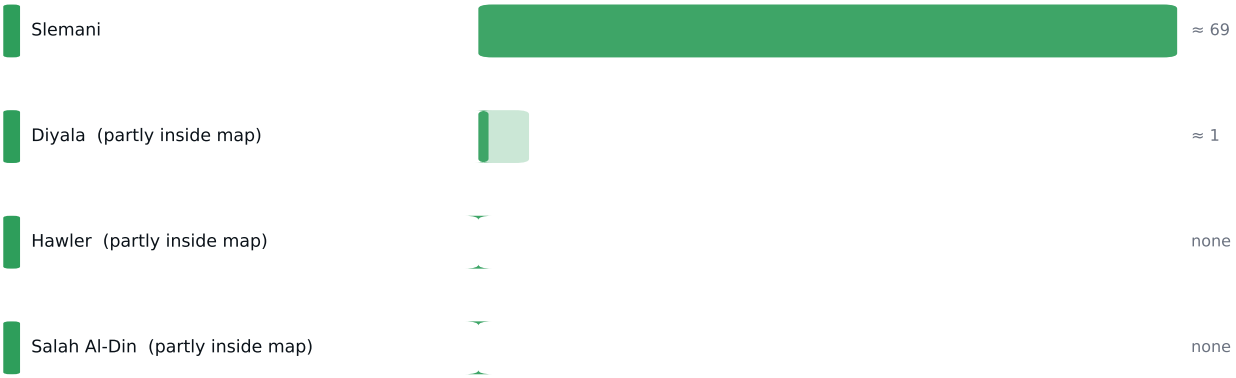


Which buildings failed

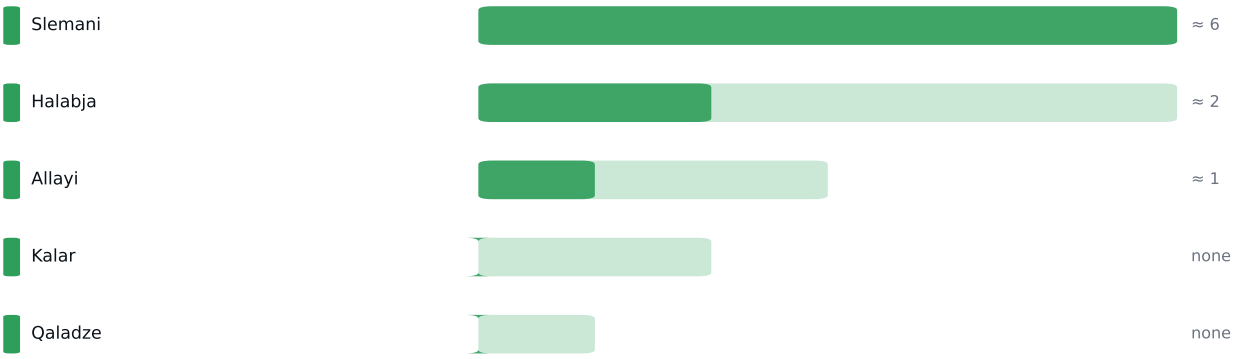
Share of each type's own buildings heavily damaged, worst first

Rubble stone, fieldstone	class A	**0.0%**	≈ 5 buildings
Adobe, earth brick	class A	**0.0%**	≈ 4 buildings
Manufactured stone with RC floors	class C	**0.0%**	≈ 20 buildings
Manufactured stone units, timber floors	class B	**0.0%**	≈ 7 buildings
Simple stone	class B	**0.0%**	≈ 4 buildings
RC frame — no / low code (ERD-L)	class C	**0.0%**	≈ 2 buildings

Governorates most affected: heavily damaged buildings, central estimate and 90% range



Cities most affected



Every governorate inside the map

Area	Buildings	Heavy (P50)	90% range	% of its stock
Slemani	575 thousand	42	18 to 290	0.01%
Diyala	42 thousand	1	none to 5	0.00%
Salah Al-Din	400	none	none to none	0%
Hawler	150	none	none to none	0%

How this report was made

Hazard engine Bumelerze SHAKEmap 26.2.0, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)

Observations 0 seismic stations, 0 felt-report cells (felt channel: none)

Damage scale IMS-25

Data sources

Built from USGS, EMSC and GEOFON earthquake parameters; building footprints from GFZ OpenBuildingMap (c) OpenStreetMap contributors, ODbL, and Microsoft, CDLA-Permissive-2.0; JRC Global Human Settlement Layer and WorldPop, CC BY 4.0; OCHA COD-AB Iraq, CC BY-IGO; GEM Global Exposure Model; IMS-25 vulnerability table. Full records, with licences and checksums: github.com/Peshawa-LH/bumelerze-engine

What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Hasan P. L. (2026) Bumelerze SHAKEmap and damage report, event bml20260260, version 2, generated 19 Oct 2026, 13:50 UTC.